Harsh Vibhuti

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EDUCATION

Washington University in St. Louis — St. Louis, MO
B.S. Mechanical Engineering, Robotics — Aug 2025 – May 2027
Engineering Coursework: Heat Transfer, Thermodynamics, Fluid Mechanics, Machine Elements, CAD, Vibrations, Machine Shop, Solid Mechanics, Material Science, Partial Differential Equations, Statistics

St. Edward's University — Austin, TX
Bachelor of Science - Mathematics (WashU Dual Degree student) — Aug 2022 – May 2025

SKILLS

Engineering & Design: 3D CAD (SolidWorks), COMSOL, Iterative Concept Prototyping, Assembly & Test Fixture Design, Finite Element Analysis (FEA), Detailed 2D Drawings (GD&T), Design for Manufacturability/Assembly (DFM/DFA), Bill of Materials (BOM) creation.
Programming & Controls: Python, MATLAB/Simulink, Pandas.
Hardware & Validation: Hands-on Fabrication, Root Cause Failure Analysis, Hardware Validation Testing, Motion Control Systems (Sensors, Actuators, Mechanisms).
Documentation: High-level Technical Writing (Specs, Test Plans, Reports, White Papers).

RELEVANT WORK EXPERIENCE

Air Quality Intern | Capital Area Council of Governments — Austin, TX | May 2025 - Aug 2025

- Automated environmental data workflows using Python, Selenium, and Pandas, reducing QA time by 95% - demonstrating systems optimization and applied problem-solving.
- Presented technical findings to city and regional stakeholders; translated engineering data into policy and operational recommendations.

A/V Specialist | St. Edward's University — Austin, TX | Jan 2023 - Aug 2024

- Engineered and calibrated complex hardware and control systems across 50+ environments, prioritizing mechanical and systems design principles to ensure optimal performance.
- Collaborated with multidisciplinary technical teams and end-users to align hardware solutions with strict operational and ergonomic requirements.

ENGINEERING PROJECTS AND EXTRACURRICULARS

WashU Design Build Fly (Avionics Subteam Member) — St. Louis, MO | Sep 2025 – Present

- Designed a custom avionics bay using 3D CAD (SolidWorks) to systematically house complex electrical components, applying Design for Assembly (DFA) principles to significantly streamline system diagnostics and troubleshooting.
- Engineered custom servo mounts and integrated servo motors (actuators) into the aircraft's motion control system, managing all wiring and controls to ensure reliable multidisciplinary performance.

WashU Rocketry (Payload Subteam member) — St. Louis, MO | Sep 2025 – Present

- Optimized payload body tube design using 3D CAD (SolidWorks), executing targeted mass-reduction geometry to yield a thinner, lighter housing with superior structural integrity validated by FEA simulations.
- Addressed high component costs by independently designing, fabricating, and assembling high-precision custom resin bearings (Form 4 SLA) to replace expensive commercial ball casters, successfully implementing a direct system cost-reduction.

Line-Following Robot (Arduino Project) — Austin, TX | March 2024

- Designed and integrated motion control systems, utilizing sensors, actuators, and mechanisms to ensure precise, autonomous path tracking.
- Troubleshot hardware issues and performed root cause failure analysis, implementing corrective design actions to sustain and improve system reliability.

LEADERSHIP AND INVOLVEMENT

Vice President: Engineers Without Borders (Washington University in St. Louis): Direct multidisciplinary engineering teams in the design and execution of sustainable water supply system, for a primary school in Bulubandi, Uganda, demonstrating proactive organizational skills to guide projects from concept to implementation.
President: Math and Physics Club (St. Edward's University): Led a STEM-focused student organization; coordinated competitions, events, and 5+ trips to undergraduate math conferences across Texas, promoting analytical thinking and team collaboration.